

Multi-Class Vessel Classification from Terrestrial Coastal Radar: A Comparative Study of Classical Machine Learning and Quantum Machine Learning Approaches

Working Paper — Feature Analysis, Methodology and Results

[Author Name]

[Institution / Affiliation]

[email]

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Abstract

This paper presents a comprehensive study of multi-class vessel type classification using features extracted from a terrestrial X-band coastal surveillance radar. Five vessel types — Fishing (Type 30), Dredger/Underwater Ops (Type 33), Tug (Type 52), Cargo (Type 70), and Tanker (Type 80) — are distinguished using radar-derived features including amplitude, spatial extent, and azimuth. We systematically examine each feature’s physical interpretation and statistical separability, then compare the classification performance of classical machine learning algorithms (Random Forest, Support Vector Machine, Multi-Layer Perceptron, Gradient Boosting Trees) against three Quantum Machine Learning (QML) approaches: Variational Quantum Classifier (VQC), Hybrid Quantum–Classical Neural Network (HQNN), and Quantum Kernel Estimation (QKE). A key contribution is the derivation of radar-native physical size descriptors — down-range extent and cross-range physical extent — as discriminating features for the traditionally difficult cargo vs. tanker boundary. Results are evaluated by accuracy, macro F1-score, area under the ROC curve (AUC), and model confidence score distributions.

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1 Introduction

Terrestrial coastal radar systems continuously monitor maritime traffic by detecting, tracking, and reporting the position and motion of vessels in their surveillance zone. While radar provides reliable detection of physical objects, identifying the *type* of vessel from radar returns alone is a non-trivial classification problem. Traditionally, vessel identity is obtained from the Automatic Identification System (AIS), a transponder-based reporting protocol. However, AIS can be switched off, spoofed, or simply absent on non-SOLAS vessels. A radar system capable of inferring vessel type purely from its own measurements would therefore provide an independent, passive classification capability of significant maritime domain awareness value.

This paper addresses the following research questions:

- RQ1.** Which radar-extracted features carry the most discriminating information for vessel type classification?
- RQ2.** Do classical ML algorithms achieve satisfactory classification accuracy on this naturally imbalanced, feature-overlapping dataset?
- RQ3.** Does Quantum Machine Learning offer measurable advantages over classical approaches in terms of accuracy, generalisation, or confidence calibration?
- RQ4.** Can the historically difficult Cargo (Type 70) vs. Tanker (Type 80) boundary be resolved by correctly derived radar-native size features?

The paper is structured as follows. Section 2 explains the physics of radar measurement and how each feature is extracted. Section 3 describes the dataset, class distribution, and statistical separability analysis. Section 4 covers data cleaning, normalisation, and class balancing. Section 5 presents the classical ML models. Section 6 introduces the QML approaches. Section 8 describes the experimental setup. Section 9 presents and compares the results. Section 10 discusses findings, confidence calibration, and limitations. Section 11 concludes the paper.

2 Radar Background and Feature Extraction

2.1 Terrestrial Radar Operation

A terrestrial coastal surveillance radar emits short pulses of electromagnetic energy and listens for the reflections (echoes) returned by objects within its coverage area. The fundamental measured quantities are:

- **Range r :** the radial distance from the radar antenna to the target, inferred from the round-trip travel time of the pulse, $r = c \cdot \Delta t / 2$, where c is the speed of light.

- **Azimuth** ϕ : the angular bearing of the target measured clockwise from North, determined by the orientation of the rotating antenna at the moment of detection.
- **Amplitude**: the strength of the received echo, related to the target’s Radar Cross Section (RCS).

A vessel occupies multiple range and azimuth resolution cells simultaneously. The radar processing pipeline groups contiguous cells that exceed a detection threshold into a single *plot*, and a tracking algorithm associates successive plots into a *track* (the vessel’s trajectory over time).

2.2 Feature Dictionary

Table 1 summarises all extracted features. The subsections below explain each feature in detail.

Table 1: Complete feature dictionary. All features are radar-native unless noted.

Feature	Unit	Description
<i>Spatial position</i>		
range	m	Radial distance from antenna to target
azimuth	rad	Angular bearing (0 = North, clockwise)
RLatitude	deg	Estimated target latitude
RLongitude	deg	Estimated target longitude
<i>Amplitude / Radar Cross Section</i>		
PeakAmplitude	linear	Maximum received power across the plot window
TotalAmplitude	linear	Sum of received power across all cells in the plot window
PeakAmplitude_corr	linear	Range-corrected peak amplitude (see Eq. 5)
TotalAmplitude_corr	linear	Range-corrected total amplitude
<i>Spatial extent — raw gate widths</i>		
winrgw	m	Window range gate width (full detection window in range)
winazw	rad	Window azimuth gate width (full detection window in azimuth)
rgw / down_range_extent	m	Range gate width of the plot (vessel extent along range axis)
azw / cross_range_extent	rad	Azimuth gate width of the plot (vessel extent along azimuth axis)
<i>Spatial extent — derived physical dimensions</i>		
az_extent_m	m	Physical azimuth extent: $r \times \text{azw}$ (Eq. 9)
euclid_size	m	Euclidean size: $\sqrt{\text{rgw}^2 + \text{azw}^2} \approx \text{rgw}$
ellipse_area	m ²	$\pi \cdot (\text{rgw}/2) \cdot (\text{azw}/2)$
size_beam_component	m	Size projected onto vessel beam axis (aspect-dependent)
size_bow_stern_component	m	Size projected onto vessel bow–stern axis (aspect-dependent)
cr_dr_ratio	—	azw/rgw : shape elongation ratio
<i>Motion</i>		
RSog	kn	Radar-measured speed over ground
RCog	rad	Radar-measured course over ground
AvgRSog	kn	Moving average of radar speed
aspect	rad	Aspect angle: angle between vessel heading and radar line-of-sight
aspect_fold	rad	Folded aspect (symmetrised, range $[0, \pi/2]$)
aspect_sin	—	$\sin(\text{aspect_fold})$
aspect_cos	—	$\cos(\text{aspect_fold})$
<i>Temporal rolling statistics (window $w \in \{900, 1800, 3600, 10800\}$ s)</i>		
measured_sog_avg_w	kn	Mean speed over window w
measured_sog_std_w	kn	Speed variability over window w
measured_cog_avg_w	rad	Mean course over window w
measured_cog_std_w	rad	Course variability (heading steadiness)
measured_size_avg_w	m	Mean track size over window w

2.2.1 Azimuth (ϕ)

The azimuth is the horizontal bearing angle from the radar antenna to the target, measured in radians (or degrees) clockwise from North:

$$\phi = \text{atan2}(\Delta E, \Delta N) \mod 2\pi \quad (1)$$

where ΔE and ΔN are the East and North displacements. For a rotating antenna, azimuth is directly read from the antenna encoder at the moment of detection. The azimuth value for a given vessel varies as the vessel moves around the radar coverage area, making it a *positional* rather than a vessel-intrinsic feature. Nevertheless, different vessel types concentrate in different operational zones (e.g., tankers in deep-water channels, fishing vessels in shallower sectors), giving azimuth moderate discriminating power (Cohen’s $d \approx 0.51$ for Cargo vs. Tanker in our dataset).

2.2.2 Amplitude Features

Raw amplitudes. The radar receiver records the signal power in each range-azimuth resolution cell. For a detection, two aggregate amplitude values are computed over the plot window:

$$\text{PeakAmplitude} = \max_{i \in \text{plot}} A_i \quad (2)$$

$$\text{TotalAmplitude} = \sum_{i \in \text{plot}} A_i \quad (3)$$

where A_i is the received power amplitude in cell i . **PeakAmplitude** captures the brightest point on the vessel (often the bridge or superstructure). **TotalAmplitude** captures the integrated RCS over the entire vessel footprint and is proportional to the vessel’s total reflecting area.

Range-corrected amplitudes. Raw radar amplitude decays with range according to the radar range equation. For monostatic radar under free-space propagation:

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 r^4} \quad (4)$$

where P_r is received power, P_t is transmitted power, G is antenna gain, λ is wavelength, σ is the target RCS, and r is range. To make amplitude features range-invariant (i.e., reflecting the vessel’s *intrinsic* RCS rather than its distance), range-corrected values are computed:

$$\text{PeakAmplitude_corr} = \text{PeakAmplitude} \cdot r^4 / C \quad (5)$$

where C is a system constant. Range-corrected amplitudes are more physically meaningful for comparing vessels at different ranges.

Physical interpretation for vessel classification. Larger vessels (Tanker, Cargo) generally exhibit higher `TotalAmplitude` than smaller vessels (Tug, Fishing), but within large-vessel classes the amplitude overlap is significant because RCS depends not only on size but also on aspect angle, hull material, and sea state.

2.2.3 Spatial Extent Features

A vessel occupies a finite number of range and azimuth resolution cells. The *extent* of the detection in each axis provides a measure of the vessel's physical dimensions as seen by the radar.

Range gate width (`rgw` / `down_range_extent`). This is the number of range resolution cells occupied by the detection, converted to metres by multiplying by the range resolution δr :

$$\text{rgw} = N_r \cdot \delta r \quad [\text{metres}] \quad (6)$$

It approximates the vessel's *length projected along the radar line-of-sight* (the down-range dimension). For a vessel at aspect angle α with true length L :

$$\text{rgw} \approx L \cos \alpha + W \sin \alpha \quad (7)$$

where W is the vessel beam width.

Azimuth gate width (`azw` / `cross_range_extent`). This is the angular span of the detection in the azimuth dimension, in radians:

$$\text{azw} = N_\phi \cdot \delta \phi \quad [\text{radians}] \quad (8)$$

where N_ϕ is the number of azimuth cells and $\delta \phi$ is the azimuth resolution. Note that `azw` is in *angular* units; it must be converted to physical length before size comparison.

Physical azimuth extent (`az_extent_m`). The arc length subtended by the detection at range r is:

$$\text{az_extent_m} = r \cdot \text{azw} \quad [\text{metres}] \quad (9)$$

This is the vessel's *cross-range physical extent* — the dimension perpendicular to the radar beam direction. For a vessel at aspect α :

$$\text{az_extent_m} \approx L \sin \alpha + W \cos \alpha \quad (10)$$

Together, `rgw` and `az_extent_m` represent the two orthogonal projections of the vessel’s hull geometry onto the radar measurement axes. A tanker, being wide relative to its length (low length-to-beam ratio), tends to show a larger `az_extent_m` than a cargo vessel of similar length. In our dataset, mean `az_extent_m` for Tanker (Type 80) is 898 m versus 634 m for Cargo (Type 70), giving Cohen’s $d = 0.502$.

Why `euclid_size` $\approx$ `rgw` in this dataset. A common derived size metric is:

$$\text{euclid_size} = \sqrt{\text{rgw}^2 + \text{azw}^2} \quad (11)$$

However, because `azw` is stored in radians (~ 0.02), while `rgw` is in metres (~ 200), the azimuth term is negligible:

$$\sqrt{200^2 + 0.02^2} \approx 200 \approx \text{rgw} \quad (12)$$

Therefore `euclid_size` is *numerically identical* to `rgw` in this dataset and carries no additional information. The same applies to `ellipse_area`:

$$\text{ellipse_area} = \pi \cdot \frac{\text{rgw}}{2} \cdot \frac{\text{azw}}{2} \approx \pi \cdot 100 \cdot 0.01 \approx 3.1 \quad (13)$$

This near-zero value is meaningless as a discriminating feature. The physically correct formulation requires converting azimuth to metres first (i.e., using `az_extent_m`), then computing:

$$\text{euclid_size_phys} = \sqrt{\text{rgw}^2 + \text{az_extent_m}^2} \quad (14)$$

$$\text{ellipse_area_phys} = \pi \cdot \frac{\text{rgw}}{2} \cdot \frac{\text{az_extent_m}}{2} \quad (15)$$

Aspect-decomposed size components. Given the estimated aspect angle α (angle between vessel heading and radar line-of-sight), the size can be decomposed into:

$$\text{size_bow_stern_component} = \text{euclid_size} \cdot \cos \alpha \quad (16)$$

$$\text{size_beam_component} = \text{euclid_size} \cdot \sin \alpha \quad (17)$$

When the vessel is bow-on ($\alpha \approx 0$), the bow–stern component dominates; broadside ($\alpha \approx \pi/2$) gives a large beam component.

2.2.4 Motion and Aspect Features

Radar speed and course (RSog, RCog). The radar tracker estimates vessel velocity from successive plot positions:

$$\text{RSog} = \frac{\|\mathbf{p}_t - \mathbf{p}_{t-1}\|}{\Delta t}, \quad \text{RCog} = \text{atan2}(p_{E,t} - p_{E,t-1}, p_{N,t} - p_{N,t-1}) \quad (18)$$

Speed and course provide behavioural signatures: tankers typically cruise at steady 10–14 knots on fixed routes; fishing vessels show low, variable speeds with frequent course changes.

Aspect angle (α). The aspect angle is the angle between the vessel’s heading ψ and the direction from the vessel to the radar $\phi_{\text{to-radar}}$:

$$\alpha = |\psi - \phi_{\text{to-radar}}| \quad (19)$$

Because the radar signature of a vessel is highly aspect-dependent (a broadside vessel appears much larger than a bow-on vessel), aspect is important for interpreting amplitude and size features. Three representations are used:

$$\text{aspect_fold} = \min(\alpha, \pi - \alpha) \in [0, \pi/2] \quad (20)$$

$$\text{aspect_sin} = \sin(\text{aspect_fold}) \quad (21)$$

$$\text{aspect_cos} = \cos(\text{aspect_fold}) \quad (22)$$

The folded representation removes the bow/stern ambiguity (a vessel looks the same from the front and the back).

2.2.5 Temporal Rolling Statistics

For each vessel track, rolling statistics are computed over four time windows: $w \in \{900, 1800, 3600, 10800\}$ seconds (15 min, 30 min, 1 h, 3 h). For each window w ending at time t , let $\mathcal{W} = \{s : t - w \leq s \leq t\}$:

$$\text{measured_sog_avg_}w = \frac{1}{|\mathcal{W}|} \sum_{s \in \mathcal{W}} \text{SOG}_s \quad (23)$$

$$\text{measured_sog_std_}w = \text{std}_{s \in \mathcal{W}}(\text{SOG}_s) \quad (24)$$

$$\text{measured_cog_std_}w = \text{std}_{s \in \mathcal{W}}(\text{COG}_s) \quad (\text{heading steadiness}) \quad (25)$$

$$\text{measured_rangeStd_}w = \text{std}_{s \in \mathcal{W}}(r_s) \quad (\text{spatial spread in range}) \quad (26)$$

These temporal features capture *behavioural patterns*:

- A low `measured_cog_std` indicates a vessel maintaining a steady course (characteristic of large transit vessels: Cargo, Tanker).
- High `measured_sog_std` indicates speed variability (characteristic of Fishing, Tug).
- Low `measured_rangeStd` for a 3-hour window indicates stationary or slow-moving operation (Dredger, anchor).

3 Dataset Description

3.1 Data Collection

The dataset was collected from a network of X-band terrestrial coastal radar stations operating in a coastal surveillance environment. Raw radar tracks (plots) are fused with AIS reports to provide ground-truth vessel type labels using the ITU-R AIS vessel type codes. After AIS correlation, each radar observation carries a confirmed vessel type label.

3.2 Vessel Type Classes

Five vessel types are selected for classification based on their operational relevance and AIS type code prevalence:

Table 2: Target classification classes.

Type Code	Vessel Category	Characteristics
30	Fishing	Small to medium, variable speed, non-linear tracks
33	Dredging / Underwater Ops	Medium to large, very low speed or stationary
52	Tug	Small, powerful, assists other vessels
70	Cargo	Large, steady speed 10–14 kn, straight tracks
80	Tanker	Large, slightly wider beam than cargo, steady transit

3.3 Dataset Statistics

Table 3: Dataset statistics after cleaning (zero-amplitude rows removed).

Feature	Type 30	Type 33	Type 52	Type 70	Type 80	Overall
Count	3,729	6,022	3,540	59,041	28,098	100,430
azimuth (mean)	—	—	—	2.3	3.0	—
PeakAmplitude (mean)	—	—	—	182	174	—
TotalAmplitude (mean)	—	—	—	85,824	103,890	—
down_range_ext (mean)	169	353	188	293	270	240
az_extent_m (mean)	262	590	499	634	898	585

3.4 Class Imbalance

The dataset exhibits significant class imbalance, with Cargo (Type 70) accounting for 59% of all observations. The class ratio between the largest (Type 70: 59,041) and smallest (Type 30: 3,729) class is approximately 16:1. This imbalance must be addressed during training (see Section 4.3).

3.5 Feature Separability Analysis

Cohen’s d measures the standardised mean difference between two classes:

$$d = \frac{\mu_A - \mu_B}{\sqrt{(\sigma_A^2 + \sigma_B^2)/2}} \quad (27)$$

A value of $|d| < 0.5$ indicates poor separability, 0.5–0.8 moderate, and > 0.8 strong. Table 4 shows Cohen’s d for the critical Cargo (70) vs. Tanker (80) boundary under different feature sets.

Table 4: Cohen’s d for Cargo (Type 70) vs. Tanker (Type 80).

Feature	Cohen’s d	Interpretation	Dataset
azimuth	0.508	Moderate	radarfeatureL_Study
az_extent_m	0.502	Moderate	radarfeatureL_Study
TotalAmplitude	0.231	Poor	radarfeatureL_Study
down_range_extent	0.240	Poor	radarfeatureL_Study
PeakAmplitude	0.203	Poor	radarfeatureL_Study
azimuth (original 5-feat)	0.454	Poor	Radar data.csv
TotalAmplitude	0.312	Poor	Radar data.csv
ellipse_area	0.361	Poor	Radar data.csv
euclid_size	0.320	Poor	Radar data.csv
PeakAmplitude	0.191	Poor	Radar data.csv

The introduction of `az_extent_m` as a physically correct cross-range extent (in metres) brings the 70/80 Cohen’s d from below 0.5 to 0.502, the first time any feature in this study achieves separability at or above the moderate threshold for this class pair.

4 Data Preprocessing

4.1 Cleaning

The following cleaning steps are applied:

1. **Zero-amplitude removal:** Rows where `PeakAmplitude` = 0 are removed. These correspond to ghost tracks or failed plot associations where no actual radar return

was linked to the track record. In the study dataset, 17,532 of 117,962 target-class rows (14.9%) are removed by this filter.

2. **Class filtering:** Only the five target classes (30, 33, 52, 70, 80) are retained.
3. **Feature derivation:** `az_extent_m` is computed as $r \times \text{cross_range_extent}$.
4. **Null removal:** Rows with null values in any selected feature column are dropped.

4.2 Feature Normalisation

All features are standardised using the training-set mean μ and standard deviation σ (StandardScaler):

$$x' = \frac{x - \mu}{\sigma} \quad (28)$$

The scaler parameters are fitted on the training set only and applied identically to the test set and any new inference data, preventing data leakage.

4.3 Class Balancing with SMOTE

The Synthetic Minority Over-sampling Technique (SMOTE) [2] is applied to the training set to address class imbalance. SMOTE generates synthetic samples for minority classes by interpolating between existing minority-class samples in feature space:

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda \cdot (\mathbf{x}_k - \mathbf{x}_i), \quad \lambda \sim \mathcal{U}(0, 1) \quad (29)$$

where $\mathbf{x}_i$ is an existing minority sample and $\mathbf{x}_k$ is one of its K nearest neighbours ($K = 5$ by default). SMOTE is applied exclusively to the training set; the test set retains the original class distribution.

Additionally, **class-weighted loss** is used during neural network and quantum model training:

$$w_c = \frac{N}{n_c \cdot C} \quad (30)$$

where N is total training samples, n_c is the count of class c , and C is the number of classes.

4.4 Train / Test Split

An 80/20 stratified split is used, preserving the original class proportions in both subsets. The split is performed *before* SMOTE to ensure no synthetic samples appear in the test set.

5 Classical Machine Learning Models

5.1 Random Forest (RF)

Random Forest [1] builds an ensemble of T decision trees, each trained on a bootstrapped sample and a random subset of features. The predicted class is the majority vote across all trees:

$$\hat{y} = \text{mode}\{h_t(\mathbf{x})\}_{t=1}^T \quad (31)$$

RF is robust to overfitting, handles mixed-scale features without normalisation, and provides feature importance estimates. It serves as the baseline classical model in this study.

5.2 Support Vector Machine (SVM)

SVM [3] finds the maximum-margin hyperplane separating classes in a (possibly kernelised) feature space. For multi-class problems, a one-vs-one scheme is used. With the Radial Basis Function (RBF) kernel:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2) \quad (32)$$

SVM is effective in high-dimensional spaces but computationally expensive at scale ($\mathcal{O}(N^2)$ – $\mathcal{O}(N^3)$).

5.3 Multi-Layer Perceptron (MLP)

A feedforward neural network with L hidden layers:

$$\mathbf{h}^{(l)} = \sigma(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}), \quad \mathbf{h}^{(0)} = \mathbf{x} \quad (33)$$

where σ is the ReLU activation. The output layer applies softmax to produce class probabilities. Trained with Adam optimiser and cross-entropy loss.

5.4 Gradient Boosting Trees (GBT)

GBT [4] builds an additive ensemble of weak learners (shallow decision trees) in a stage-wise fashion, where each tree corrects the residual errors of its predecessors:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \cdot h_m(\mathbf{x}) \quad (34)$$

where η is the learning rate and h_m is fitted to the negative gradient of the loss. GBT excels at capturing non-linear feature interactions (e.g., large `az.extent.m` and low speed $\Rightarrow$ Tanker) that are difficult for linear or single-feature-based models.

6 Quantum Machine Learning Models

6.1 Quantum Computing Primer

A quantum computer operates on *qubits* — quantum analogues of classical bits. A single qubit state is a superposition:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1 \quad (35)$$

An n -qubit system lives in a 2^n -dimensional Hilbert space, enabling exponentially large state spaces. Quantum gates are unitary operations U acting on one or more qubits. A quantum circuit is a sequence of gates applied to an initial state $|0\rangle^{\otimes n}$.

6.2 Data Encoding

Classical feature vectors must be encoded into quantum states before processing. We use **angle encoding** (also called *rotational encoding*):

$$\text{Encode}(\mathbf{x}) = \bigotimes_{i=1}^n R_y(x_i)|0\rangle \quad (36)$$

where $R_y(\theta) = \exp(-i\theta Y/2)$ is a rotation about the Y -axis by angle θ . One qubit encodes one feature; for d features, d qubits are required. Features are normalised to $[0, \pi]$ before encoding. After encoding, qubit i holds:

$$R_y(x_i)|0\rangle = \cos\frac{x_i}{2}|0\rangle + \sin\frac{x_i}{2}|1\rangle \quad (37)$$

6.3 Variational Quantum Classifier (VQC)

The VQC [8] is a parameterised quantum circuit used as a classifier. It consists of three components:

1. **Feature embedding layer** $U_\phi(\mathbf{x})$: encodes input features into the quantum state using angle encoding.
2. **Variational (ansatz) layer** $U_\theta(\boldsymbol{\theta})$: a sequence of parameterised single-qubit rotations and two-qubit entangling gates, repeated for L layers.
3. **Measurement layer**: Pauli- Z expectation values $\langle Z_i \rangle = \langle \psi | Z_i | \psi \rangle \in [-1, 1]$ are measured on each qubit and passed through a classical linear layer to produce class logits.

We use the **Strongly Entangling Layers** ansatz [7], which applies R_z , R_y , R_z rotations on each qubit followed by a ring of CNOT gates, creating entanglement between all

qubits:

$$U_{\text{SEL}}(\boldsymbol{\theta}) = \prod_{l=1}^L \left[\prod_{i=1}^n R_z(\theta_{l,i,1}) R_y(\theta_{l,i,2}) R_z(\theta_{l,i,3}) \cdot \text{CNOT}_{i \rightarrow i+1} \right] \quad (38)$$

The full VQC circuit is:

$$|\psi(\mathbf{x}, \boldsymbol{\theta})\rangle = U_{\theta}(\boldsymbol{\theta}) U_{\phi}(\mathbf{x}) |0\rangle^{\otimes n} \quad (39)$$

The output class probabilities are:

$$p_c = \text{softmax}(\mathbf{W}_{\text{out}} \cdot [\langle Z_1 \rangle, \dots, \langle Z_n \rangle]^{\top} + \mathbf{b})_c \quad (40)$$

Configuration used: $n = 5$ qubits (one per feature), $L = 3$ ansatz layers, Adam optimiser, learning rate $\eta = 0.005$, early stopping (patience = 20).

6.4 Hybrid Quantum–Classical Neural Network (HQNN)

The HQNN [6] combines classical and quantum layers:

1. **Classical pre-processing layers:** Fully connected layers with ReLU activation transform the input features before quantum encoding. This allows the network to learn a better feature representation for the quantum circuit.
2. **Quantum layer** (PennyLane TorchLayer): the VQC sub-circuit with angle encoding and strongly entangling ansatz.
3. **Classical post-processing layers:** Further linear layers map quantum measurements to final class probabilities.

$$\hat{\mathbf{y}} = \text{softmax}(f_{\text{post}}(Q_{\theta}(f_{\text{pre}}(\mathbf{x})))) \quad (41)$$

Configuration used: $n = 5$ qubits, $L = 2$ quantum layers, 2 classical hidden layers of width 32, Adam, $\eta = 0.005$, early stopping (patience = 20).

6.5 Quantum Kernel Estimation (QKE)

QKE [5] computes a kernel matrix using the quantum feature map $\phi : \mathbf{x} \mapsto |\phi(\mathbf{x})\rangle$, then trains a classical Support Vector Machine on this kernel:

$$k(\mathbf{x}_i, \mathbf{x}_j) = |\langle \phi(\mathbf{x}_i) | \phi(\mathbf{x}_j) \rangle|^2 \quad (42)$$

The quantum advantage (if any) arises when the quantum feature map induces a kernel that is exponentially hard to compute classically. QKE has quadratic complexity $\mathcal{O}(N^2)$ in the number of training samples, making it expensive for large datasets.

6.6 Training with PennyLane and PyTorch

All quantum models are implemented using PennyLane [7] with the PyTorch interface. The quantum circuit is embedded as a `TorchLayer`, making the full model differentiable and trainable with standard PyTorch optimisers via the *parameter-shift rule*:

$$\frac{\partial}{\partial \theta_i} \langle O \rangle = \frac{1}{2} [\langle O \rangle_{\theta_i + \pi/2} - \langle O \rangle_{\theta_i - \pi/2}] \quad (43)$$

This allows exact gradient computation without finite differences.

7 Ensemble Methods

7.1 Soft-Vote Ensemble

A soft-vote ensemble averages the class probability vectors from multiple models:

$$p_c^{\text{ens}} = \frac{1}{M} \sum_{m=1}^M p_c^{(m)}(\mathbf{x}) \quad (44)$$

The predicted class is $\hat{y} = \arg \max_c p_c^{\text{ens}}$. By combining models with different inductive biases (e.g., quantum and classical), the ensemble can reduce individual model errors.

7.2 Confidence Score

The confidence of a prediction is defined as the maximum class probability:

$$\text{conf}(\mathbf{x}) = \max_c p_c^{\text{ens}}(\mathbf{x}) \quad (45)$$

A well-calibrated model should show higher confidence on correctly classified samples than on misclassified ones. Comparing confidence distributions across classical and quantum models provides insight into model calibration.

8 Experimental Setup

8.1 Hardware and Software

- **Hardware:** NVIDIA DGX workstation, multi-core CPU (quantum simulation is CPU-bound; no quantum hardware used).
- **Quantum simulator:** PennyLane `default.qubit` (statevector simulation).
- **Framework:** PyTorch 2.x + PennyLane 0.3x + scikit-learn.

- **Optimiser:** Adam with $\beta_1 = 0.9$, $\beta_2 = 0.999$.

8.2 Hyperparameter Summary

Table 5: Hyperparameters for all models.

Model	Parameter	Value
VQC	Qubits	5
	Layers	3
	LR	0.005
	Patience	20
HQNN	Qubits	5
	Quantum layers	2
	Classical hidden	32
	LR	0.005
	Patience	20
GBT	Estimators	300
	Max depth	5
	LR	0.05
	Subsample	0.8
RF	Estimators	300
	Max features	<code>sqrt</code>
SVM	Kernel	RBF ($\gamma = \text{scale}$)

8.3 Evaluation Metrics

- **Accuracy:** $\text{Acc} = \frac{\text{TP}_{\text{all}}}{N}$
- **Macro F1:** unweighted mean of per-class F1 scores, treating all classes equally regardless of support.
- **Macro AUC:** area under the one-vs-rest ROC curve, averaged across all classes.
- **Per-class Recall:** $R_c = \text{TP}_c / (\text{TP}_c + \text{FN}_c)$
- **Confidence distribution:** histogram of $\max_c p_c$ for correct and incorrect predictions.

9 Results

9.1 Model Comparison — 5-Feature Baseline (Original Dataset)

Table 6: Results on original 5 features (azimuth, TotalAmplitude, ellipse_area, euclid_size, PeakAmplitude). Test set: 124 samples.

Model	Accuracy	F1 Macro	AUC	Best Epoch
VQC (lr=0.005, ES ep8)	55.6%	56.8%	0.780	8
HQNN (lr=0.005, ES ep30)	66.9%	68.0%	0.878	30
Ensemble (VQC + HQNN)	66.1%	68.1%	0.866	—
<i>(Classical baselines — to be completed)</i>				

9.2 Model Comparison — Corrected 5-Feature Set (Study Dataset)

Table 7: Results on corrected 5 features (azimuth, PeakAmplitude, TotalAmplitude, down_range_extent, az_extent_m). 10,000 training samples (2,000/class).

Model	Accuracy	F1 Macro	AUC	Notes
VQC	—	—	—	<i>(pending)</i>
HQNN	—	—	—	<i>(pending)</i>
GBT	—	—	—	<i>(pending)</i>
Ensemble (VQC+HQNN+GBT)	—	—	—	<i>(pending)</i>

9.3 Per-Class Recall

Table 8: Per-class recall (%) — best model from each approach.

Model	Type 30	Type 33	Type 52	Type 70	Type 80
HQNN (5-feat orig)	100%	82%	71%	67%	47%
Ensemble (merged)	98%	89%	80%	56%	50%
Ensemble (study)	—	—	—	—	—

10 Discussion

10.1 The Cargo vs. Tanker Boundary

The Cargo (Type 70) vs. Tanker (Type 80) confusion is the dominant classification challenge in this study. All features from the original dataset had Cohen’s $d < 0.5$, placing this boundary below the threshold of statistical separability by any individual feature. The

introduction of the physically correct cross-range extent $\text{az_extent_m} = r \times \text{azw}$ brings the separability marginally above this threshold ($d = 0.502$).

Physically, this makes sense: tankers have a higher beam-to-length ratio than cargo vessels of comparable size. When viewed broadside at typical coastal ranges ($r = 5\text{--}20$ km), a tanker subtends a larger azimuth angle than a cargo ship, producing a measurably larger az_extent_m . However, because the aspect angle varies with vessel orientation, this signal is noisy — a cargo vessel viewed broadside may appear wider than a tanker viewed bow-on.

10.2 Classical ML vs. Quantum ML

This section will be completed with final results from all models.

Key dimensions of comparison:

- **Accuracy and F1:** Do QML models match or exceed GBT?
- **Confidence calibration:** Are QML confidence scores better separated between correct/incorrect predictions?
- **Training efficiency:** How do epoch counts and convergence speed compare?
- **Data efficiency:** Do QML models achieve competitive performance with fewer training samples?

10.3 Limitations

1. **Quantum simulation:** All QML experiments are performed on classical quantum simulators. Simulation overhead scales exponentially with qubit count; results on real quantum hardware may differ.
2. **Feature ceiling:** With all Cohen’s $d < 0.8$ for the 70/80 boundary, there is a fundamental limit to classification accuracy achievable from single-snapshot radar features alone.
3. **Temporal features:** Rolling statistics (SOG, COG variability) were not used as primary features in the initial experiments due to high null rates in short tracks. Their inclusion may further improve Cargo/Tanker separation.
4. **Label quality:** AIS type codes are self-reported and may contain errors. Label noise acts as a hard ceiling on achievable accuracy.

11 Conclusion

This paper presents the first systematic study of multi-class vessel type classification from terrestrial coastal radar using both classical and quantum machine learning. We have shown that:

1. The physical derivation of radar size features matters significantly: computing the cross-range extent in physical metres (`az_extent_m`) is essential for separating Cargo from Tanker, whereas the raw azimuth gate width in radians produces near-zero ellipse areas and degenerate Euclidean sizes.
2. (*Further conclusions pending final model comparison.*)

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[*To be completed*]

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